Table 21.5 Decay heat from unpartitioned fuel, fission products and heavy elements

	Decay heat (W/kg initial U)		
Cooling time	**Total**	**Fission products**	**Heavy elements**
1 d	54.3	38.7	15.6
90 d	16.2	13.0	3.27
180 d	11.6	9.20	2.42
1 y	7.67	5.93	1.73
5 y	2.98	1.70	1.28
10 y	2.17	1.12	1.05
100 y	0.28	0.016	0.27
10^3 y	0.13	0.000036	0.13
10^4 y	0.017	0.000034	0.017
10^5 y	0.0080	0.000020	0.0080
10^6 y	0.0021	0.0000007	0.0021

Limits: Fission products 28<Z<68, heavy element Z>79.
(Basic data as in Table 21.2)

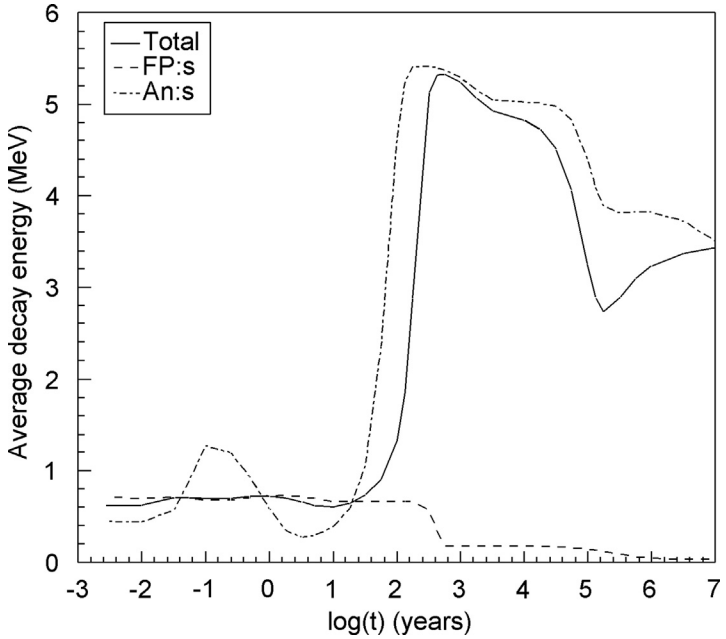


Figure 21.9 Average energy per decay in spent PWR fuel after 33 MWd/kg burnup.

21.4. MANAGEMENT OF SPENT FUEL

There are in principle several options for the handling of spent fuel. It can be:

a) Stored temporarily in pools, or dry storage, awaiting a decision on its future treatment

b) Reprocessed on order to recover fissionable material like U and Pu

c) Placed in a deep underground geologic repository
d) Ejected into space (not very likely at present)
Figure 21.11 shows the known and predicted total amounts of spent fuel, reprocessed fuel, and stored fuel as a function of time.

21.4.1. Transport of spent reactor fuel

The storage capacity of reactor pools is normally several years' production but can be increased by adding neutron absorbers to the storage racks. Eventually the fuel assemblies must be transferred in special transport flasks to (interim) storage sites, sent for reprocessing, or sent to final disposal.

The loading of used fuel the assemblies in the transport flask requires shielding and remote handling, and the heat continual cooling. Therefore, almost all operations are carried out under water. Because each transport is expensive the transport flasks are designed to carry several assemblies of different types. A 30 t (gross weight) flask may carry 4 PWR or 9 BWR assemblies (~1 t U), a 100 t flask ~12 PWR or ~30 BWR assemblies (~6 t U). The inner cavity of such flasks is normally surrounded by a neutron absorbing shield; see Figure 21.10. They have shock absorbers, and sometimes cooling fins on the outside. A filled 100 t flask with 1 year old fuel develops ~60 kW heat; the

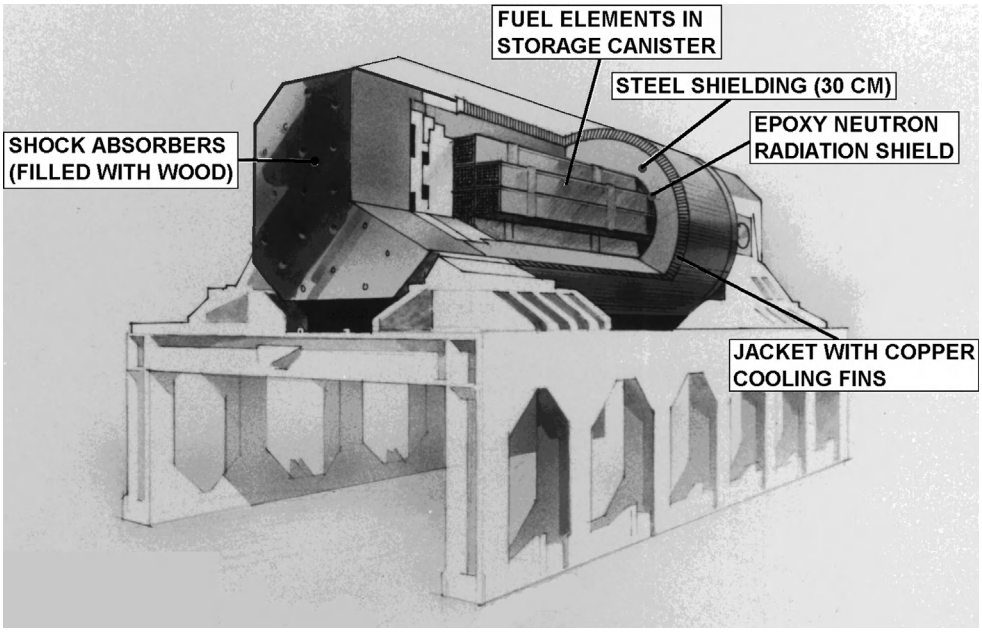


Figure 21.10 Transport cask TN-17 for 7 PWR or 17 BWR fuel elements; 1.96 m, length 6.15 m, empty weight 76 tons (SKB, Sweden).

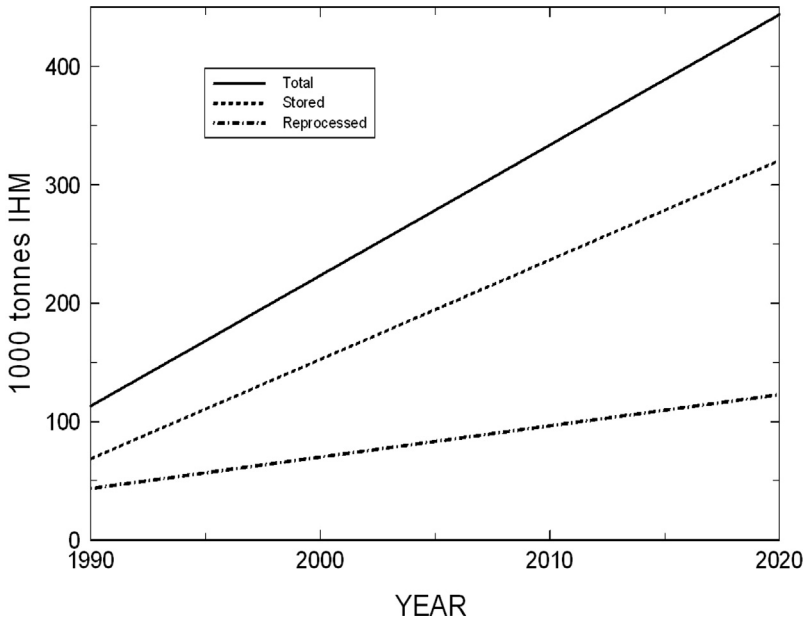


Figure 21.11 Total amounts of spent, stored, and reprocessed nuclear fuels as predicted by IAEA 2010.

design cooling capacity of the flask is ~100 kW. The cavity in some flasks is filled with water because the flasks are loaded and unloaded under water, and the water functions as heat conducting and neutron absorbing material (several actinides decay by spontaneous fission; α,n-reactions occur with light target atoms). Figure 21.10 shows a fuel cask for dry transport of 7 PWR or 17 BWR assemblies by special trucks, by rail, or by sea in special RO-RO type ships.

The flasks are designed for exceedingly severe treatment: free fall from 9 m onto a concrete floor, 30 min gasoline fire (~800°C), submersion into 15 m of water, etc., without being damaged. After loading a flask the outside of the flask must be decontaminated; often it is quite difficult to remove all activity, and a removable plastic covering is used. Before unloading, water filled flasks are flushed to remove any activity leaked from the fuel or suspended crud, which is present on the used fuel elements from BWRs and most PWRs. Further decontamination is carried out after the flask has been emptied.

Dry air filled flasks are used in many countries. They are either cooled by filling with water before unloading (the escaping steam is collected and condensed) or unloaded hot and dry by remote handling. In the former case, further handling is the same as for water filled flasks.

Shielded transport is also required for solidified high level waste, hulls, plutonium containing material, and for some intermediate and α-active waste. Special containers are

used for each type. As an example, in the United Kingdom plutonium containers made of wood and cadmium are limited to carrying 10 kg Pu; the container weighs 175 kg, is 1.3 m high, and 0.8 m in diameter.

21.4.2. Interim storage facilities

In the once-through fuel cycle, or with a limited capacity for reprocessing, separate spent fuel element interim storage facilities become compulsory. These facilities are either of the wet pool type (Czech Republic, Finland, Germany, Japan, Sweden, United Kingdom) or of the dry vault type (France, Germany, United Kingdom). The pool types consist of large water filled basins containing a geometrically safe (with regard to criticality) lattice of stainless steel racks for the spent fuel elements. The water is circulated for cooling. In general such pools are either located in a building at ground level or underground at 30–50 m depth. To increase pool capacity, boron steel plates are sometimes inserted thereby permitting denser packing of fuel elements without any criticality problem.

Zircaloy clad oxide fuel elements can be stored for decades in storage pools with very little risk of leakage. Metal fuels, especially those canned in magnesium or aluminum alloys, are less resistant and should not be stored as such in this manner for a prolonged time. The corrosion resistance of aluminum or magnesium clad fuel can be improved by electrolytic treatment yielding a protective oxide layer.

Some of the stringent requirements on the storage pools are: $k_{\text{eff}} < 0.95$ (even if unused fuel elements are introduced), earthquake safety, no possible water loss, water level automatically kept constant, adequate leakage and radiation monitoring systems, water temperature $<65^{\circ}$C, acceptably low radiation level in working areas, etc.

Dry vault storage may also be used. The fuel elements are stacked horizontally or vertically in concrete pipes which allows cooling of the fuel elements by air convection (forced or natural).

For long term storage ($\geq$10 y) the fuel elements must usually be recanned. For this purpose single fuel elements or bundles are placed in cylindrical containers, and the void is filled with some suitable material like lead, which has good heat conductivity and also provides some radiation protection. Depending on the external condition at the final storing place (humidity, temperature, etc.) the canisters are surrounded by an additional container to improve the lifetime of the fuel elements, which – preferably – should not be exposed to the biosphere until all radioactivity has disappeared.

21.5. ALTERNATIVE FUEL CYCLES

Fission energy can be obtained from uranium, using the *uranium once-through* option and the *uranium-plutonium fuel cycle*, and from thorium, by the *thorium-uranium fuel cycle*. Each fuel cycle offers a number of alternative routes with respect to reactor type, reprocessing,